

# Improving Accuracy in Retrieval-Augmented Question Answering Systems

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## Abstract

Retrieval-augmented generation (RAG) systems combine a retrieval component with a language model to answer questions grounded in external documents. This paper evaluates three retrieval strategies — fixed-size chunking, semantic chunking, and structure-aware chunking — and finds that semantic chunking improves answer accuracy by 12% over fixed-size chunking on a legal-document benchmark, while structure-aware chunking performs best on clause-level precision tasks.

# 1. Introduction

Large language models are prone to hallucination when asked questions that require specific factual grounding in a source document. Retrieval-augmented generation addresses this by retrieving relevant passages before generation, allowing the model to answer using only supplied context. This paper investigates how the choice of chunking strategy affects downstream answer accuracy and citation correctness.

## 2. Methodology

We evaluated three chunking approaches across two benchmark sets: a general-domain financial report corpus and a legal-contract corpus (CUAD). For each approach, we measured retrieval precision at  $k=3$  and end-to-end answer correctness using a set of 150 human-annotated question-answer pairs, each labeled with the expected source page.

### 3. Results

Semantic chunking achieved 89% retrieval precision at  $k=3$  on the financial corpus, compared to 77% for fixed-size chunking. On the legal corpus, structure-aware chunking achieved 94% clause-level precision, substantially outperforming both alternatives. These results suggest that chunking strategy should be selected based on document type rather than applied uniformly.

## 4. Conclusion

Matching chunking strategy to document structure meaningfully improves both retrieval precision and the reliability of page-level citations in RAG systems. Future work should explore automatic detection of document type to select the optimal chunking strategy without manual configuration.